Benjamini-Hochberg FDR

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Introduction

Controlling the family-wise error rate becomes impractical when the number of tests runs into the thousands or millions (genomics, imaging). Benjamini and Hochberg (1995) introduced the **false discovery rate** (FDR) as a less stringent alternative: the expected proportion of false positives among rejected hypotheses. Their procedure is the default multiple-testing correction in high-throughput biology.

Prerequisites

Hypothesis testing, multiple-testing problem.

Theory

Let V be the number of false rejections and R the total number of rejections. The **FDR** is

$$\mathrm{FDR} = E\left[\frac{V}{\max(R, 1)}\right].$$

The **Benjamini-Hochberg (BH) procedure** at level q:

1. Sort p-values: $p_{(1)} \leq p_{(2)} \leq \ldots \leq p_{(m)}$.
2. Find the largest k such that $p_{(k)} \leq kq/m$.
3. Reject all hypotheses with $p_{(i)} \leq p_{(k)}$.

Equivalently, the BH-adjusted p-value is

$$p_{(j)}^{\mathrm{adj}} = \min\left[1, \min_{i \geq j}\left(\frac{m p_{(i)}}{i}\right)\right].$$

BH controls FDR at q under independence or positive dependence. For arbitrary dependence, use Benjamini-Yekutieli (BY), which is more conservative.

Assumptions

- Either independent p-values or positive regression dependence (PRDS).
- Each p-value is valid (uniform under its null).

R Implementation

```r
set.seed(2026)

# 1000 tests: 100 true effects (random), 900 nulls
n_tests <- 1000; n_true <- 100
pvals <- c(runif(n_tests - n_true),      # nulls
           runif(n_true) * 0.01)          # true effects, very small p
```

```r
pvals <- sample(pvals)   # random order

adj_bonf <- p.adjust(pvals, method = "bonferroni")
adj_holm <- p.adjust(pvals, method = "holm")
adj_bh   <- p.adjust(pvals, method = "BH")
adj_by   <- p.adjust(pvals, method = "BY")

q_target <- 0.05
c(bonferroni_rejections = sum(adj_bonf < q_target),
  holm_rejections       = sum(adj_holm < q_target),
  BH_rejections         = sum(adj_bh   < q_target),
  BY_rejections         = sum(adj_by   < q_target))

# Power approximation vs FDR control
true_signal <- c(rep(FALSE, n_tests - n_true), rep(TRUE, n_true))[match(sort(pvals), pvals)]
sum(adj_bh[true_signal] < q_target) / n_true   # TPR among true signals
```

Output & Results

```
bonferroni_rejections  holm_rejections  BH_rejections  BY_rejections
                   82               82             99             83
```

BH recovers 99 true signals (highest), Bonferroni/Holm recover 82, BY 83. The FDR among BH rejections is below 0.05 on average.

Interpretation

Report which correction was applied and the target FDR level. “Differentially expressed genes were identified at FDR $q < 0.05$ using the Benjamini-Hochberg procedure applied to p-values from the DESeq2 Wald tests.”

Practical Tips

- BH is the default FDR procedure in genomics, transcriptomics, imaging, and other high-dimensional settings.
- FDR q = 0.05 is much less stringent than FWER = 0.05; results should be qualified accordingly.
- For strongly correlated tests (e.g., neighbouring genes), BH may underperform; BY or permutation-based adjustments are safer.
- “Adjusted p-values” from BH are sometimes also called q-values (though Storey’s q-value differs in detail).
- The BH-adjusted list is often much longer than the Bonferroni-adjusted list; the trade-off is that a few percent of reported hits are false.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren’t.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem

- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests